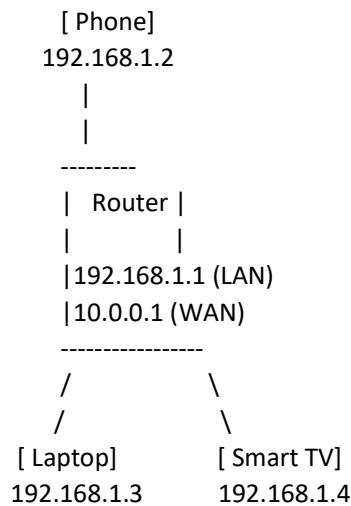


IP ROUTING AND ROUTING PROTOCOL

1. Simple Network Diagram (with IP Addresses)



All devices are in same LAN: **192.168.1.0/24**

2. Packet Travel (Laptop → Friend's Computer in Another Subnet)

Example

- Your laptop: **192.168.1.3**
- Friend PC: **192.168.2.5**
- Router:
 - LAN: 192.168.1.1
 - WAN: 192.168.2.1

Step-by-Step

1. Laptop checks IP → 192.168.2.5 (different subnet)
2. Sends packet to **default gateway (192.168.1.1)**
3. Router receives packet
4. Router checks routing table:
192.168.1.0/24 → LAN interface
192.168.2.0/24 → WAN interface
5. Router forwards packet to **192.168.2.0 network**
6. Friend's computer receives packet

Final Path

Laptop → Router → Friend PC

3. Routing Table (Zomato-Style Delivery Network)

Destination Network	Subnet Mask	Next Hop	Interface
10.1.0.0	255.255.255.0	Direct	Delhi
10.2.0.0	255.255.255.0	10.1.0.2	Delhi
10.3.0.0	255.255.255.0	10.1.0.3	Mumbai

Meaning

- Delhi, Mumbai, Bangalore = different delivery hubs
- Router decides best path like assigning delivery driver

4. Static vs Dynamic Routing

Static Routing

Pros

- Simple to set up
- No extra network traffic
- More secure

Cons

- Manual updates needed
- Not scalable (bad for big apps like Zomato)
- Fails if link breaks

Dynamic Routing (RIP/OSPF)

Pros

- Automatically updates routes
- Good for large systems (like Uber/Zomato)
- Handles failures automatically

Cons

- More complex
- Uses CPU/memory
- Can cause overhead traffic

5. Simple OSPF Configuration + Explanation

Sample Config

```
router ospf 1
```

```
network 192.168.1.0 0.0.0.255 area 0
```

```
network 192.168.2.0 0.0.0.255 area 0
```

Explanation

- router ospf 1 → Starts OSPF process number 1
- network 192.168.1.0... → Includes first network in OSPF
- 0.0.0.255 → Wildcard mask (means /24 subnet)
- area 0 → Backbone area of OSPF (main routing area)
- Second line → adds another network to OSPF